

High-Level System Architecture

Autonomous Delivery Cart

1. Mobile Layer (Customer App – Youssef)

- **Platform:** iOS/Android (built with Flutter/React Native).
- **Responsibilities:**
 - User registration/login (via AWS Cognito).
 - Browsing supermarket/pharmacy inventory (via API Gateway → Lambda → DynamoDB/RDS).
 - Placing orders.
 - Receiving notifications.
 - Tracking cart.
 - Unlocking cart (via SMS code verification).

2. Vendor Layer (Web Portal – Ahmed)

- **Platform:** Web App.
- **Responsibilities:**
 - Vendor login (Cognito).
 - Inventory management (API Gateway → Lambda → DB).
 - Receiving and preparing orders.
 - Triggering cart dispatch.
 - Monitoring order status.

3. Manager Layer (Dashboard – Ahmed)

- **Platform:** AWS.
- **Responsibilities:**
 - Monitoring all carts and orders.
 - Viewing analytics.
 - Exporting reports.
 - Managing vendors.

4. Cart / Edge Device Layer (Jetson Nano/Tx2 + Sensors)

- **Components:**
 - Sensors: LiDAR, ZED stereo camera, Azure Kinect DK.
 - Communication: MQTT over AWS IoT Core.
 - Control: ROS (Robot Operating System) running on Jetson.
- **Responsibilities:**
 - Send telemetry (GPS, battery, sensor data) every 30s.
 - Receive delivery commands.
 - Unlock storage via backend confirmation (Cognito + verification code).
 - Return to garage after delivery.

5. Cloud Backend (AWS – Ahmed)

Core Services:

- **API Gateway** → Entry point for mobile/web apps.
- **AWS Lambda** → Handles business logic (orders, inventory, cart commands).
- **AWS IoT Core (MQTT)** → Real-time cart ↔ cloud communication.
- **Amazon Cognito** → Authentication (customers, vendors, managers).
- **Amazon RDS** → Structured data (orders, users, inventory).
- **Amazon DynamoDB** → Semi-structured logs, telemetry.
- **Amazon S3 + Glacier** → Sensor/video storage (raw + archive).
- **Amazon SNS / Pinpoint** → Notifications + SMS code delivery.
- **Amazon QuickSight / Athena** → Analytics dashboards for managers.
- **CloudWatch + X-Ray** → Monitoring & logging.

6. Data Flow (End-to-End Example)

1. **Customer places order** → Mobile app → API Gateway → Lambda → DB.
2. **Vendor sees order** → Vendor portal → Confirms & loads cart.
3. **Vendor dispatches cart** → API Gateway → IoT Core → Cart receives command.
4. **Cart moves & updates location** → IoT Core → Lambda → DB → Customer sees real-time tracking.
5. **Cart arrives** → Customer gets SMS code (SNS).
6. **Customer unlocks cart** → App → API Gateway → IoT Core → Cart unlocks.
7. **Cart returns** → IoT Core → Lambda → Status updated.
8. **Manager dashboard** → Queries RDS/DynamoDB + Analytics pipeline for reports.

System Architecture

